American Computer Science League

2025-2026 ● Contest 3: Short Problems Solutions ● Intermediate Division

1. Boolean Algebra

$$
\begin{aligned}
& A(\overline{B + C}) + \overline{B\overline{C}} + C \\
&= A(\overline{B} * \overline{C}) + \overline{B} + C + C \\
&= A * \overline{B} * \overline{C} + \overline{B} + C \\
&= \overline{B}(A\overline{C} + 1) + C \\
&= \overline{B} + C
\end{aligned}
$$

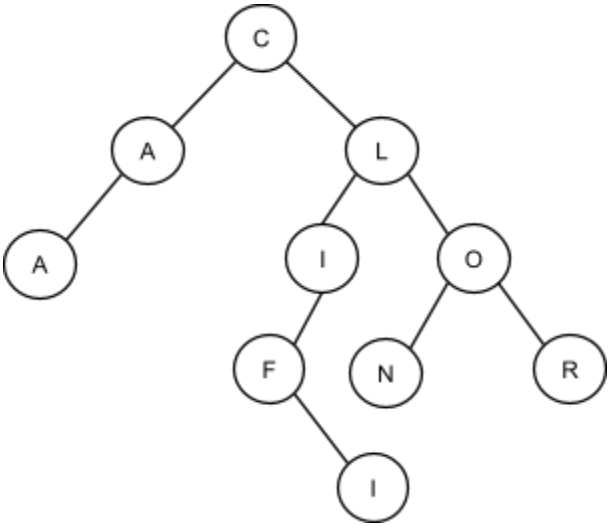
B. $\overline{B} + C$

2. Boolean Algebra

$$
\begin{aligned}
& \overline{A(\overline{B + C}) + \overline{B + \overline{C}}} \\
&= \overline{A(\overline{B + C})} * (B + \overline{C}) \\
&= (\overline{A} + B + C) * (B + \overline{C}) \\
&= \overline{A}B + \overline{A} * \overline{C} + B + B\overline{C} + BC + C\overline{C} \\
&= \overline{A} * \overline{C} + B
\end{aligned}
$$

The general ordered triples using asterisk notation (0,*,0) and (*,1,*) yields 5 ordered triples: (0,0,0), (0,1,0), (1,1,1), (0,1,1), (1,1,0)

B. 5

3. Data Structures The binary search tree for **CALIFORNIA** is:  The internal path length is the sum of the depths of all of the nodes which is 2*1+3*2+3*3+1*4=21	D. 21
4. Data Structures The operations on an initially empty queue that is FIFO is as follows: S, SA, SAC, AC, ACR, ACRA, CRA, RA, RAM, RAME, AME, AMEN, AMENT, AMENTO, MENTO Therefore, the next item popped from the front of the queue is M.	E. M
5. FSAs The direct translation of this FSA to a regular expression is: 100*101*1. There are no other choices that are equivalent expressions to this one.	E. 100*101*1

6. FSAs and Regular Expressions Given the regular expression **`[a-ct-v][e-m|o-u]*[aeiou]`** the words are evaluated as follows: violin - stops at n cello - valid trombone - stops at b clarinet - stops at r trumpet - stops at last t oboe - stops at o saxophone - stops at s tuba - stops at b flute - stops at f There is only 1 that is accepted.	A. 1